Pharmaceutical Reform and Pharmacy Entry

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County-level Empirical Setting

- Unit of analysis: county-by-year panel, 2012–2022.
- Main controlled sample: 2012–2019, with population, nighttime lights, and road-density controls.
- Main treatment indicator:

$$Reform_{ct} = \mathbf{1}\{t \geq PolicyYear_c\}.$$

- Baseline FE specification:

$$Y_{ct} = \beta_0 + \beta_1 Reform_{ct} + \beta X_{ct} + \alpha_c + \lambda_t + \varepsilon_{ct}.$$

- Controls include log population, log nighttime lights, and log road density where available.
- Standard errors are clustered at the county level.

County Sample and Descriptive Statistics

Variable	N	Mean	SD	P25	Median	P75
Pharmacy count	17,507	97.12	109.80	26	64	131
Log pharmacy count	15,984	4.16	1.15	3.56	4.29	4.94
Annual change in count	14,804	19.25	50.94	0	9	27
Annual change in log count	13,277	0.25	0.65	0.00	0.13	0.29
Has pharmacy	17,507	0.91	0.28	1	1	1
Chain pharmacy count	17,507	8.08	27.11	0	0	1
Non-chain pharmacy count	17,507	89.04	97.88	25	60	121
DID	17,507	0.65	0.48	0	1	1

County TWFE DID: Pharmacy Stock and Annual Change

	Count	Count + controls	Annual Δ count
Post policy	1.672	0.128	20.030***
	(1.986)	(1.877)	(2.019)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	No	Yes	Yes
N	17,463	17,463	14,714
R^2	0.820	0.841	0.333

TWFE DID: Log Count and Extensive Margin

	ln(count)	Has pharmacy
Post policy	-0.001	0.034***
	(0.017)	(0.004)
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
N	15,940	17,463
R^2	0.876	0.662

Continuous DID

	Count	ln(count)
Post policy $\times$ ln(public hospitals in 2012 + 1)	17.901***	-0.081***
	(3.811)	(0.024)
County FE	Yes	Yes
Year FE	Yes	Yes
Controls	Yes	Yes
N	17,463	15,940
R^2	0.843	0.877

CSDID and PPML

	Pharmacy count	ln(count)
ATT	-2.178	0.010
	(2.077)	(0.032)
Pre-average	-2.403	0.038
	(2.696)	(0.077)
Post-average	10.002*	0.084
	(4.610)	(0.084)
Controls	Yes	Yes
Control group	Not-yet-treated	Not-yet-treated

	PPML count FE
Post policy	-0.023
	(0.015)
County FE	Yes
Year FE	Yes
Controls	Yes
N	17,449

County Distance to Nearest Hospital

	Mean km	Share $\leq$2km	$\leq$2km count	$>$2km count
Post policy	0.006	0.000	3.898***	-3.770**
	(0.121)	(0.005)	(1.084)	(1.269)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	15,940	15,940	17,463	17,463
R^2	0.666	0.742	0.848	0.760

Distance measures are based on each pharmacy's nearest hospital of any ownership type.

County Pharmacies Near Public and Private Hospitals

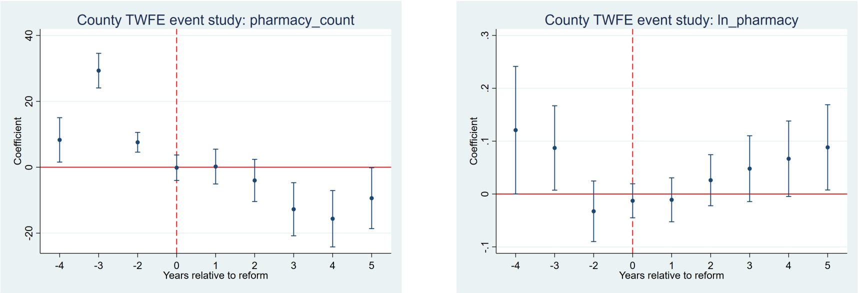
	Public $\leq$2km	Private $\leq$2km	Public share	Private share
Post policy	4.234***	8.888***	-0.015*	0.012*
	(1.067)	(1.110)	(0.007)	(0.005)
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
N	15,940	15,940	15,940	15,940
R^2	0.781	0.869	0.704	0.846

Public/private 2km counts are not mutually exclusive: a pharmacy can be within 2km of both a public and a private hospital.

County Chain vs Non-chain Outcomes

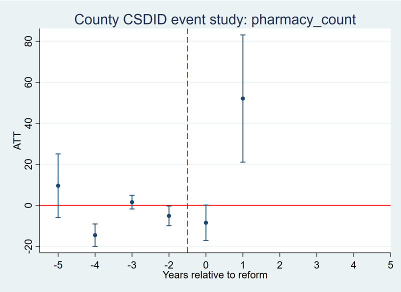
	Chain count	Non-chain count	Chain share
Post policy	2.069***	-1.942	0.007***
	(0.350)	(1.707)	(0.002)
County FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Controls	Yes	Yes	Yes
N	17,463	17,463	17,463
R^2	0.641	0.823	0.661

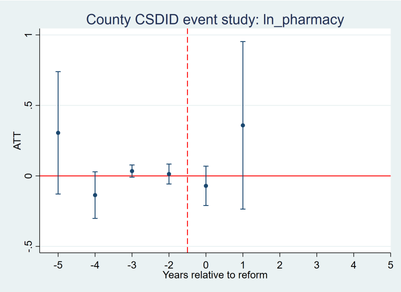
County TWFE Event-study Diagnostics



Standard TWFE event-study graphs with county and year fixed effects. Relative year -1 is omitted.

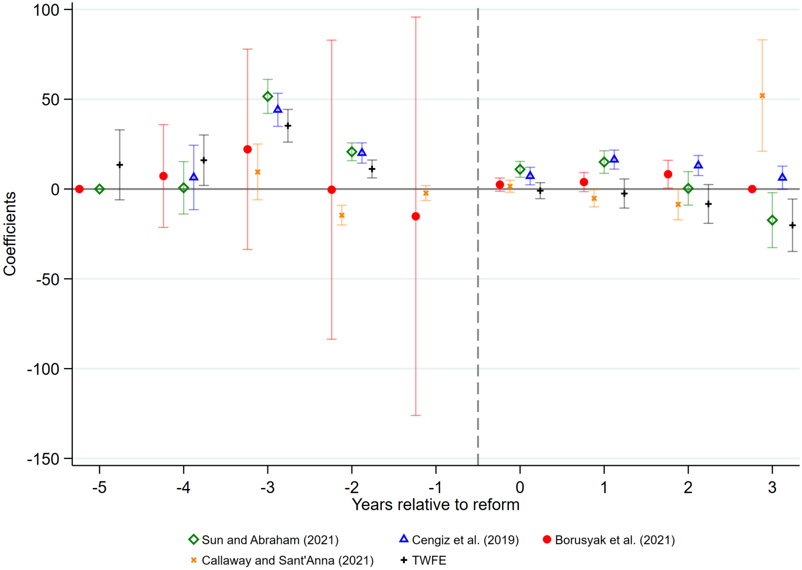
County CSDID Event-study Diagnostics



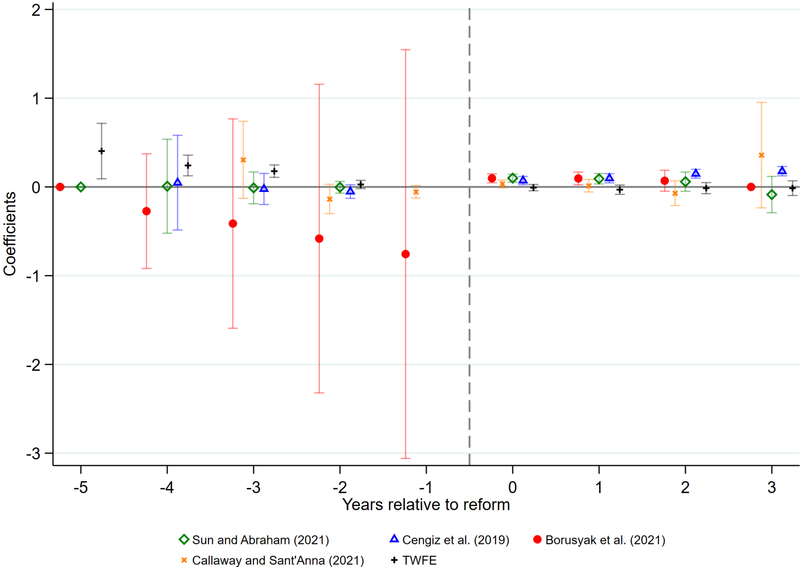


CSDID uses not-yet-treated counties as controls. The updated event window is $[-5, +5]$.

County Five-estimator Event-study Robustness: Count



County Five-estimator Event-study Robustness: ln(count)



Hospital-level DDD Specification

- Unit of analysis: hospital-by-year panel with pharmacy outcomes measured within 1km, 2km, and 3km buffers.
- Treatment contrasts public hospitals with non-public hospitals after local reform implementation:

$$Y_{ict} = \beta_0 + \beta(Public_i \times Post_{ct}) + \theta Post_{ct} + X'_{ict}\gamma + \alpha_i + \lambda_t + \varepsilon_{ict}.$$

- β is the differential post-reform change around public hospitals.
- $\theta + \beta$ is the post-reform change around public hospitals relative to the pre-period.
- Controls include nighttime lights around hospitals where available. Standard errors are clustered at the county level.

Hospital DDD: Pharmacy Count

	1km	2km	3km
Public × Post	-0.403*	-1.272**	-2.148*
	(0.192)	(0.491)	(0.856)
Post policy	1.081***	4.171***	8.757***
	(0.148)	(0.430)	(0.814)
Hospital FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
NTL control	Yes	Yes	Yes
N	164,674	164,674	164,674
R^2	0.895	0.906	0.911

The public-hospital interaction is negative, while the post-policy level shift is positive.

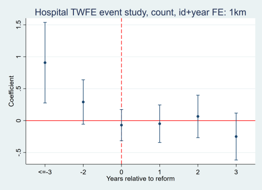
Hospital Log and PPML Outcomes

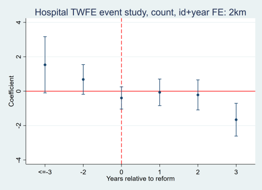
	ln(count)			PPML count		
	1km	2km	3km	1km	2km	3km
Public × Post	-0.005	0.025	0.018	-0.036*	-0.029*	-0.027*
	(0.017)	(0.017)	(0.017)	(0.015)	(0.013)	(0.012)
Post policy	0.012	0.013	0.007	–	–	–
	(0.015)	(0.015)	(0.016)	–	–	–
Hospital FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
NTL control	Yes	Yes	Yes	Yes	Yes	Yes
N	134,788	147,008	151,443	147,711	156,544	159,753
R^2	0.913	0.938	0.944	–	–	–

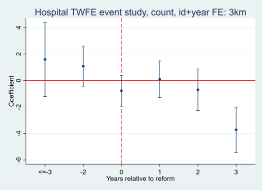
Hospital CSDID Average ATT

	1km	2km	3km
Count	-0.004	-0.447	-1.127
	(0.169)	(0.433)	(0.755)
ln(count)	-0.009	0.004	-0.006
	(0.017)	(0.016)	(0.015)

Hospital TWFE Event Study: Count, FE

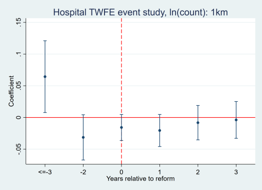


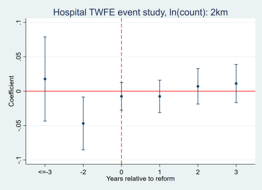


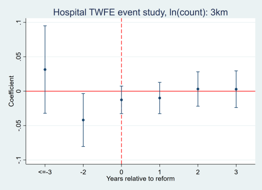


Window is restricted to event time $[-3, 3]$; relative year -1 is omitted. Models absorb hospital and year fixed effects.

Hospital TWFE Event Study: ln(count), FE







ln(count+1) event-study graphs are omitted. Models absorb hospital and year fixed effects.

How the County and Hospital Results Connect

- **County-level estimates show market expansion.** Pharmacy counts rise after reform in the county FE models, and annual count changes increase sharply.
- **Distance results do not imply broad movement away from hospitals.** Average distance and the within-2km share are almost unchanged, while counts within 2km rise.
- **Public-hospital proximity rises in absolute terms but weakens in relative terms.** Pharmacies within 2km of public hospitals increase, but the public-hospital share falls while the private-hospital share rises.
- **Hospital DDD is a relative comparison.** The negative Public $\times$ Post coefficient means public hospitals experience smaller growth than non-public hospitals, not necessarily an absolute decline around public hospitals.
- **Interpretation.** Reform appears to expand the county pharmacy market overall, with growth near hospitals, but new entry is relatively less concentrated around public hospitals than around private or non-public hospital areas.